

ASSIGNMENT #3

Course: CHE 301. CHEMICAL ENGINEERING THERMODYNAMICS
Term: Spring 2020
Professor: Dr. Michael Caracotsios
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Your name: _____

Due Date: Wednesday February 19 by 5:00pm in my office

Number of points.....100

Your score.....

Note: All problems are worth 100 points. Your final score will be the average value

Problem 3A: Review of the Ideal Gas and Maxwell Equations

Prove that:

$$\rho C_p \left(\frac{\partial T}{\partial P} \right)_H = \left(\frac{\partial \ln V}{\partial \ln T} \right)_P - 1$$

Using the formula you derived calculate the temperature change of an ideal gas that is undergoing an adiabatic expansion through a throttling valve.

Problem 3B: Review of the Ideal Gas and Maxwell Equations

Prove that:

$$C_v \left(\frac{\partial P}{\partial \rho} \right)_s = C_p \left(\frac{\partial P}{\partial \rho} \right)_T$$

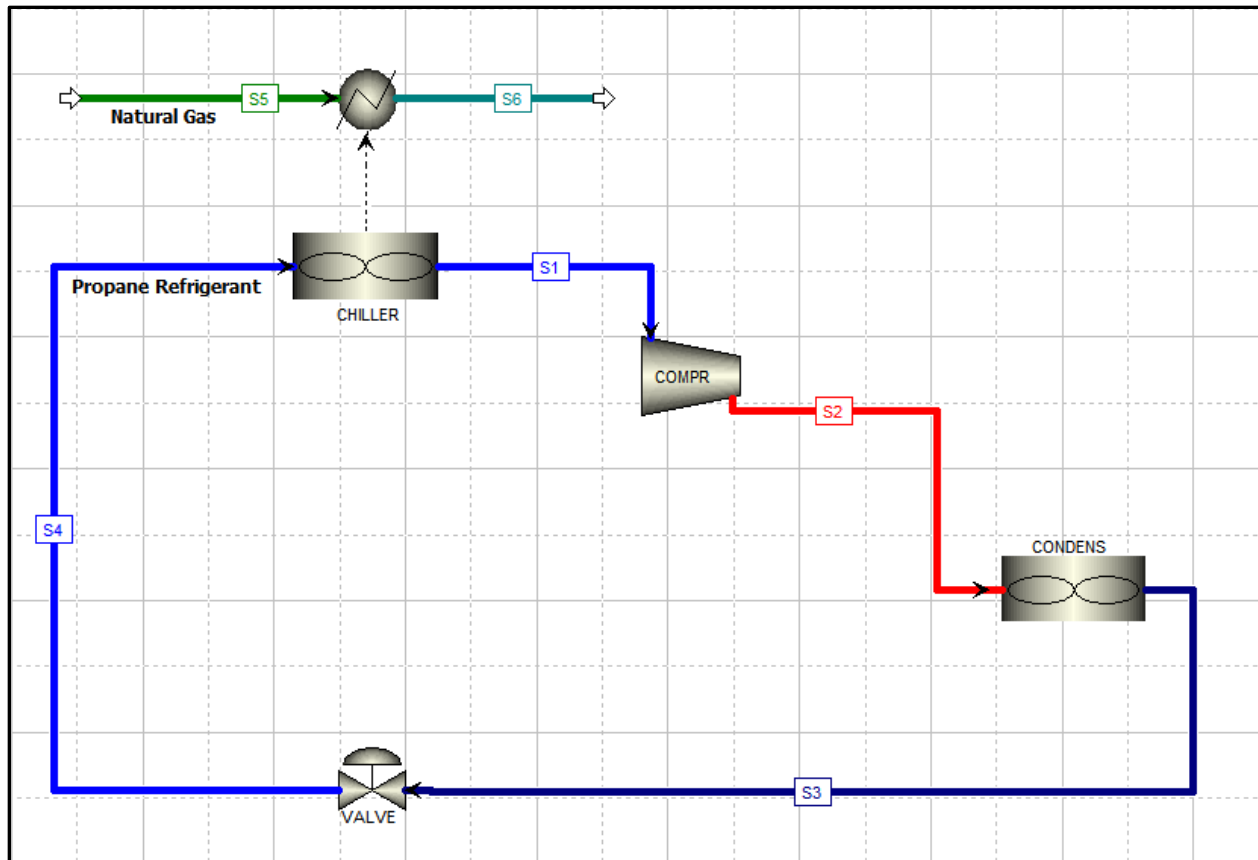
The adiabatic speed of sound is defined as $v^2 = \left(\frac{\partial P}{\partial \rho} \right)_s$. Using the formula you derived above

calculate the adiabatic speed of sound in the air at ambient temperature assuming that the air behaves as an ideal gas. Compare your result with values published in the open literature.

Problem 3C: Natural Gas Cooling using Refrigerant Propane

We wish to cool $\{100 \text{ MMSCFD}\}$ of natural gas that consists primarily of methane from $\{25^\circ\text{C to } -37^\circ\text{C}\}$ using propane as the refrigerant. A process flowsheet diagram for this refrigeration system is shown below. Saturated liquid propane at $\{35^\circ\text{C}\}$ is expanded in a throttling valve down to a pressure of $\{1\text{bar}\}$ and it is directed to a chiller, where the cooling of the natural gas stream takes place. The refrigerant from the chiller is compressed and subsequently cooled in the condenser using cooling water. Using the **CHE301 Calculator** and assuming that propane and methane behave as ideal gases calculate the following:

1. The amount of refrigerant required in KMTA (assume a 90% operating factor)
2. The duties (in MW) of the chiller and the condenser.
3. The power requirement (in MW) of the compressor assuming
 - a. 75% isentropic efficiency
 - b. 75% polytropic efficiency
4. The COP (Coefficient of Performance) of the refrigeration cycle.
5. Compare your COP with the maximum possible value for this refrigeration cycle



Place all your answers in boxes for easy readability. Clearly state your assumptions and references for additional data you may need to answer the questions.